Confidence interval for difference of two means, dependent samples

Weight loss example, kg

Background The 365 team has developed a diet and an exercise program for losing weight. It seems that it works like a charm.
However, you are interested in how much weight are you likely to lose.
You have a sample of 10 people who have already completed the 12-week program.

Task 1 Calculate the mean and standard deviation of the dataset

Task 2 Determine the appropriate statistic to use

Task 3 Calculate the 95% confidence interval

Task 4 Interpret the result

Optional You can try to calculate the 90% and 99% confidence intervals to see the difference. There is no solution provided for these cases.

Subject	Weight before (kg)	Weight after (kg)	Difference
1	103.68	92.87	-10.81
2	110.68	101.58	-9.10
3	119.05	105.66	-13.39
4	101.75	96.18	-5.57
5	91.69	86.97	-4.72
6	112.03	105.90	-6.13
7	88.84	80.56	-8.28
8	105.18	97.00	-8.18
9	110.37	99.27	-11.10
10	120.99	107.44	-13.55

Mean		-9.08
St. Dev		3.11
Statistic	T-test	
95% Confidence Interval		0.025
(10-1) degrees of freedom@0.025		2.262

T	CI low	CI high
95%	-11.31	-6.86

2. The Math Formula (using the t-distribution)

$$\mathrm{CI} = \bar{x} \pm \left(t^* \cdot \frac{s}{\sqrt{n}}\right)$$

- $\bar{x}$ (x-bar): The **mean** (average) of your sample.
- t^*: The **critical value**. This comes from a t-table based on how confident you want to be (like 95%) and your "degrees of freedom."
- s: The **standard deviation** of your sample (how spread out the numbers are).
- n: The **sample size** (how many items or people you measured).
- $\frac{s}{\sqrt{n}}$: This whole part is called the **Standard Error**.